STATS 201A Lecture 12

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$$Y = X\beta + \underline{\varepsilon} \to Z(x)$$

$$Y_D = \begin{pmatrix} y(x_1) \\ \vdots \\ y(x_n) \end{pmatrix}_{n\times 1} = \left(f_j(x_i)\right)_{n\times k} \begin{pmatrix} \beta_1 \\ \vdots \\ \beta_k \end{pmatrix}_{k\times 1} + \begin{pmatrix} Z(x_1) \\ \vdots \\ Z(x_n) \end{pmatrix}$$

Let $\tilde{Y}_D = R^{-\frac{1}{2}} Y_D$, $\tilde{F} = R^{-\frac{1}{2}} F$, $\tilde{Z} = R^{-\frac{1}{2}} Z$
$\tilde{F}^\intercal \tilde{F} = F^\intercal R^{-1} F$
$\tilde{Y}_D = \tilde{F}\beta + \tilde{Z}$
$\operatorname{Cov}(\tilde{Z}) = R^{-\frac{1}{2}} \operatorname{Cov}(Z) R^{-\frac{1}{2}} = \sigma^2 \mathbb{I}_{n\times n}$
Least Squares: $\hat{\beta} = (\tilde{F}^\intercal \tilde{F})^{-1} \tilde{F}^\intercal \tilde{Y}_D = (F^\intercal R^{-1} F)^{-1} F^\intercal R^{-1} Y_D$

$$\hat{Y} = \begin{pmatrix} \hat{y}(x_1) \\ \vdots \\ \hat{y}(x_n) \end{pmatrix} = F\hat{\beta} + \underbrace{\begin{pmatrix} \nu(x_1, x_i) & \cdots & \nu(x_n, x_i) \end{pmatrix}}_{\nu(x_i)^\intercal} \times R^{-1}(Y_D - F\hat{\beta}) = F\hat{\beta} + RR^{-1}(Y_D - F\hat{\beta}) = Y_D$$

$$y = \beta_0 + \beta_1 x + \cdots + \beta_R x^R + \varepsilon$$

$$y = f(x) + \varepsilon, \qquad \varepsilon \sim N(0, \tau^2)$$

Kirging model: $Y = \mu + Z(x)$